Design Zero-shot, One-shot, and Few-shot Prompts to Summarize an Article into 50 Words and Compare Accuracy, Completeness, and Readability

Aim

To design zero-shot, one-shot, and few-shot prompts for producing a 50-word article summary and compare the resulting summaries for factual accuracy, coverage of essential information, and readability.

Input: Sample Article

Note: The question did not include an article. The following fictional article is used only to demonstrate the experiment.

A university introduced a smart fitness watch programme for 200 student volunteers over eight weeks. The watches tracked daily steps, heart rate, and sleep duration. Students received personalised activity reminders and weekly progress reports. Average daily steps increased from 5,000 to 7,000, while average sleep duration increased by 30 minutes. Participants said reminders helped them maintain regular exercise routines. However, some students reported inaccurate readings during intense workouts and concerns about the privacy of their health data. The university stated that participation was voluntary and that collected data would be encrypted. Researchers noted that the study had no control group and that longer studies were needed before drawing firm conclusions about health benefits.

Algorithm

1. Read the article and identify its main findings, numerical facts, concerns, and limitations.

2. Write a common instruction requiring exactly 50 words and no unsupported claims.

3. Create a zero-shot prompt with no examples, a one-shot prompt with one example, and a few-shot prompt with two examples.

4. Submit each prompt to the same language model with the same settings.

5. Count each response using len(summary.split()). If it is not 50 words, request a revision and recount.

6. Compare accuracy against the article, completeness against key facts, and readability through clear wording and sentence flow.

Common Instruction

Summarise the target article in exactly 50 words. Preserve the main findings, important numbers, concerns, and study limitations. Use clear, neutral language. Do not add unsupported facts. Return only the summary. Count words by whitespace.

Zero-shot Prompt

Use the common instruction followed by:
Target article: [Paste the sample article.]
No example is supplied.

For a fair comparison, only the number of examples changes between prompts.

One-shot Prompt

Use the common instruction, the following example, and then the target article.

Example article: A college library extended evening hours for one month. Visits increased, but staffing costs rose. The trial covered only examination season.
Example summary: A college library tested longer evening hours for one month during examinations. Visits increased, suggesting students valued additional access. However, staffing costs also rose. Because the trial covered only examination season, its findings may not reflect normal demand. Further evaluation is needed before deciding whether to retain the extended schedule.

Target article: [Paste the sample article.]

Few-shot Prompt

Use the common instruction and Example 1 above. Add the following second example, then the target article.

Example article: A school tested reusable lunch boxes for six weeks. Packaging waste fell by 20 percent. Some pupils forgot their boxes. Only one school participated.
Example summary: A school tested reusable lunch boxes for six weeks and reported a 20 percent reduction in packaging waste. Some pupils forgot to bring their boxes, creating practical difficulties. Because only one school participated, the results may not apply elsewhere. More trials are needed before recommending wider adoption of the scheme.

Target article: [Paste the sample article.]

Both example summaries contain 50 words under the whitespace counting rule. Their purpose is to demonstrate factual, balanced summaries that include limitations.

Python Program

Python 3; no external packages required. Replace ARTICLE_TEXT, INSTRUCTION_TEXT, EXAMPLE_ONE, and EXAMPLE_TWO with the article, common instruction, and complete examples printed above. Use triple-quoted strings for these four constants.

```
# Offline prompt-design and validation demonstration.
# Sample responses are supplied below; no LLM is called.
article = ARTICLE_TEXT
instruction = INSTRUCTION_TEXT
example1 = EXAMPLE_ONE
example2 = EXAMPLE_TWO
prompts = {
    "Zero-shot": instruction,
    "One-shot": instruction + "\n" + example1,
    "Few-shot": instruction + "\n" + example1
                + "\n" + example2
}
# Use each printed prompt in the same language model.
for method, prompt in prompts.items():
    print("\n" + method + " PROMPT")
    print(prompt + "\nTarget article:\n" + article)
# Paste the three model responses, in the same order.
for method in prompts:
    summary = input(method + " summary: ").strip()
    count = len(summary.split())
    print(method, "word count:", count)
    if count != 50:
        print("Revise the response to exactly 50 words.")
    else:
        print(summary)
# Manually assess factual accuracy, coverage, and clarity.
```

The program constructs prompts and checks pasted responses. It does not automatically call a language model or score factual accuracy.

Output: Illustrative Summaries

These are prepared example responses, not results from an external model run. All three were entered into the Python program to verify the word counts.

Zero-shot Summary - 50 Words

Over eight weeks, a university fitness watch programme helped students increase daily steps and sleep duration. Reminders and progress reports supported exercise routines. Some participants raised concerns about inaccurate workout readings and health data privacy. Researchers recommended longer studies because the programme lacked a control group to confirm health benefits.

One-shot Summary - 50 Words

A university tested fitness watches with 200 students for eight weeks. Daily steps rose from 5,000 to 7,000, and sleep increased by 30 minutes. Reminders encouraged exercise, but inaccurate readings and privacy concerns emerged. Participation was voluntary and data encryption promised. No control group existed; longer studies were still needed.

Few-shot Summary - 50 Words

In an eight week university study, 200 students used fitness watches. Daily steps increased from 5,000 to 7,000; sleep rose by 30 minutes. Reminders supported exercise. Reading inaccuracies and privacy concerns remained despite promised encryption and voluntary participation. Without a control group, researchers urged longer studies before confirming health benefits.

Verified Console Output

Zero-shot word count: 50
One-shot word count: 50
Few-shot word count: 50

Comparison of the Sample Summaries

Accuracy

Zero-shot: Mostly faithful, but "helped students increase" implies causation that the uncontrolled study cannot establish. One-shot: Accurately retains the numerical findings and describes encryption as promised. Few-shot: Accurately reports observed changes and explicitly avoids confirming health benefits.

Completeness

Zero-shot: Covers the trend, reminders, concerns, and lack of a control group, but omits the participant count, exact improvements, voluntary participation, and encryption. One-shot: Includes these major details, although weekly reports are omitted. Few-shot: Also includes the main details and the cautious health conclusion; weekly reports are omitted.

Readability

Zero-shot: Simple and smooth. One-shot: Clear, with logical ordering of findings and concerns. Few-shot: Concise and coherent, although the semicolon and compressed final sentences make it slightly denser.

Result

Three prompting formats were designed, and all illustrative summaries passed the 50-word check. For these samples, the few-shot summary best preserves the study limitation and cautious conclusion; the one-shot summary is similarly complete and easy to read. This example does not prove that few-shot prompting always performs better. Actual performance must be assessed using responses from a controlled model run.